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# PAPER\_720: Doc 43: Universal Permanence Equation, AGN Feedback, and the Final Parsec Problem

**Author:** Daniel T. Murphy **Date:** 2025

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## Abstract

Doc 43 (UFE ORB EXP 2\_24\_07Mar2025) presents the Universal Permanence (UP) equation, a master UQFF sum including all field contributions ( $U_{gi}$ ,  $U_m$ , TRZ zone, QS factor), applied to three astronomical phenomena: the Red Dwarf plasma orb (Drawing 30, Bearden TRZ), AGN feedback mass-metallicity relation (Drawing 31, Sanchez et al.), and the Final Parsec problem for SMBH binaries. The non-local jump probability  $P \approx 0.49/\text{s}$  demonstrates emergent quantum jumps in the UQFF framework.

## 1. Universal Permanence Equation

$$\begin{aligned}
 \text{UP}(t) &= \sum_i k_i U_{gi} + \sum_j \frac{\mu_j}{r_j} (1 - e^{-\gamma t^-} \cos(\pi t_n)) \hat{\phi}_j U_{mj} + T_s^{\mu\nu} + U_b + NN + QS + ACE + DCE + [SSq] + IF + QV \\
 &\approx 1.03 \times 10^{20} \cdot QS
 \end{aligned}$$

where:

$$t^- = -t_n \cdot e^{\pi - t_n} \quad (t_n = 13.68 \text{ s} \implies t^- \approx -3.75 \times 10^{-4} \text{ s})$$

## 2. Non-Local Jump Probability

$$P = 1 - e^{-\gamma |t^-|} \approx 0.490 \text{ s}^{-1}$$

with  $\gamma = 10^3 \text{ s}^{-1}$ . Observed: 1-1.5 non-local jumps per video frame.

## 3. Energy Density

$$\rho_{react}(t) = 10^{15} \cdot e^{-0.001 \cdot t_n} \approx 9.86 \times 10^{14} \text{ W/m}^3$$

## 4. Metal Retention Factor $f_Z$

| Galaxy Type        | $f_Z$ |
|--------------------|-------|
| Over-massive SMBH  | 0.89  |
| Under-massive SMBH | 0.85  |

| Galaxy Type       | $f_Z$ |
|-------------------|-------|
| Star-forming disk | 0.73  |
| Quenched disk     | 0.51  |

$$f_{CGM} = \frac{M_{CGM}}{M_{vir}}$$

## 5. Final Parsec

$$U_{g^4}^{FP}(t) = k_4 \cdot \rho_{SCm} \cdot \frac{M_{BH}^{binary}}{d_{parsec}} \cdot e^{-\alpha t} \cdot \cos(\pi t_n)$$

$$M_{BH}^{binary} = 8.15 \times 10^{36} \text{ kg}, d_{parsec} = 2.55 \times 10^{20} \text{ m:}$$

$$U_{g^4}^{FP}(0) \approx 7.64 \times 10^{-6} \text{ J/m}^3$$

## 6. AGN Feedback

$$U_{g^4}^{AGN}(t) \approx 8.40 \times 10^{-6} \cdot e^{-0.001t} \cdot \cos(\pi t_n) \text{ J/m}^3$$

for  $\Delta M_{BH} = 1 \text{ dex}$ ,  $f_{feedback} \approx 0.1$ .

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{\{U\_Bi\_i\}}$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M-\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{Edd}^{UQFF} = L_{Edd} \cdot \left( 1 + \frac{\rho_{SCm} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{Edd} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{SCm} = 7.09 \times 10^{-37} \text{ kg/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{jet}^{UQFF} = P_{BZ} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{crit}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{SCm} \cdot t)$  modulates jet power at the phonon frequency.

**M- $\sigma$  correction (PAPER\_1048):** The phonon-corrected M- $\sigma$  relation becomes  $M_{BH} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{SCm} / \omega_{bulge})$ .

## Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

*Upgrade from PAPER\_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER\_1041 (Cool-Core Buoyancy Balance), and PAPER\_1079 (Cooling-Flow Suppression). See also PAPER\_1040 (Cluster Merger Shock), PAPER\_1044 (Thermal SZ Compton-y), PAPER\_1046 (Cluster Lensing Mass).*

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left( 1 + \left( \frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

### Hydrostatic mass bias reduction (PAPER\_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes  $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$  at cluster cores, partially resolving the Planck SZ–CMB mass tension.

**Cool-core stabilization (PAPER\_1041/1079):** AGN feedback couples to the SCm buoyancy field via  $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$ , suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

**Phonon frequency coupling:**  $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$  sets the temporal scale for buoyancy oscillations; the ratio  $\omega_{\text{SCm}}/\omega_{\text{sound}}$  governs the phonon transmission efficiency across the ICM.

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector    | Domain             | Late-Corpus Result                               |
|-----------|--------------------|--------------------------------------------------|
| 1 (EH)    | General Relativity | Canonical Einstein-Hilbert                       |
| 2 (YM)    | Yang-Mills gauge   | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac) | Fermion / LENR     | Kozima neutron-drop (PAPER_1061)                 |

| Sector     | Domain                      | Late-Corpus Result                           |
|------------|-----------------------------|----------------------------------------------|
| 4 (SCm)    | Superconducting manifold    | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical   |
| 5 (Mag)    | Um magnetism                | Heaviside amplifier (PAPER_1072)             |
| 6 (Buoy)   | $F_{\{U\_Bi\_i\}}$ buoyancy | Variational EOM (PAPER_1065)                 |
| 7 (Aether) | Vacuum background           | Two-component $\rho$ (PAPER_1051)            |
| 8 (LENR)   | Nuclear transmutation       | COP parametric (PAPER_1081)                  |
| 9 (KK)     | Kaluza-Klein 26D            | $S_{26}^{(3)}$ compactification (PAPER_1080) |

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{BH}})(\partial^\mu \phi_{\text{BH}}) - V(\phi_{\text{BH}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{BH}}) = \frac{1}{2}m^2\phi_{\text{BH}}^2 + \frac{\lambda}{4!}\phi_{\text{BH}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{BH}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{BH}}} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \rho_{\text{vac},[\text{SCm}]}g_{\mu\nu} + F_{U\_Bi\_i}/r^2 = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{BH}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( - \exp! \left( - \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.183 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This

threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 2, \quad n_{\text{channel}} = 19/26$$

Since  $p_{\text{DVP}} = 2$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M\_BH/M\_{M\_sun} yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[\text{SSq}] \cdot m/M_{\odot}}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                        | Status                       |
|----------------|----------------------------------------------|-----------------------------------|------------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.183           | PASS<br>Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 2$              | PASS<br>Sub-threshold        |
| BSH layers     | 26 harmonic terms                            | $j = 1 \dots 26, \cos(2\pi j/26)$ | PASS Full 26D<br>projection  |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS<br>exponential     | PASS Canonical               |
| [SSq]          | 0.57                                         | Applied in BSH<br>saturation      | PASS Canonical               |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                                                | SM / Experiment                    | Source                              | Alignment          |
|----------------------------------|------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                                               | 1/137.036                          | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{m}^{-2}$<br>$2(UQFFvacuumterm) 1.114 \times 10^{-52} \text{m}^{-2}$ | Planck 2018                        | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression                                  | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | $F_{U\_Bi\_i}$ unique gravitational correction                                                 | Not yet measured                   | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U\_Bi\_i}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.*

## References

- UQFF Doc 43 (UFE\_ORB\_EXP\_2\_24\_07Mar2025.docx)
- grok\_share\_ba508f76c8e.txt entry #69
- Sanchez et al. AGN feedback mass-metallicity relation
- Session 176, v5.33

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by update\_{corpus\\_crossrefs}.py (Session 225, April 2026).*

| Paper      | Title                                           |
|------------|-------------------------------------------------|
| PAPER_1002 | AGN Buoyancy-Corrected Eddington Luminosity     |
| PAPER_1009 | 3C273 AGN $F_{U\_Bi\_i}$ Jet Modulation         |
| PAPER_1010 | TON618 AGN $F_{U\_Bi\_i}$ Jet Modulation        |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching |
| PAPER_1048 | M-Sigma Phonon-Corrected Relation               |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback     |

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1072 | SCm Activation Function Phonon Threshold         |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy      |
| PAPER_1068 | Wolfram Physics Bridge WSTP Symbolic Export      |

10 cross-reference(s) identified.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                     | Purpose                                    | Key Result                                                        |
|--------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| <code>fneutron_{s26_coupling}.py</code>    | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS                              |
| <code>kozima_{scm_cross_section}.py</code> | Sym-modulated neutron-drop cross-section   | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}]^n/26)$ |
| <code>kozima_{wstp_kernel}.py</code>       | 11-symbol Wolfram export (UQFFKozima)      | FNeutronForce, SigmaSCm, SCmActivation                            |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \cdot \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}} \cdot (F_{\{U, Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp[-(\omega - \omega_{\text{SCm}})^2/(2 \cdot \Gamma^2)] \cdot (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                                  | Purpose                                       | Key Result                |
|-----------------------------------------|-----------------------------------------------|---------------------------|
| <code>ramanujan_{polylog_s26}.py</code> | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| <code>s26_{wstp\_kernel}.py</code>      | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \cdot Li_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module                           | Purpose                                | Key Result                     |
|----------------------------------|----------------------------------------|--------------------------------|
| <code>mock_{theta_q26}.py</code> | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer $(a; q)_n$ |

**Core equations:**  $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q; q)_n$  -  $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2; q^2)_n$  -  $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q; q^2)_n$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                                      | Purpose                                    | Key Result                                    |
|---------------------------------------------|--------------------------------------------|-----------------------------------------------|
| <code>ramanujan_{pi_uqff}.py</code>         | Classical + UQFF-modified<br>1/pi + 26D    | 21 digits classical, 15 UQFF, 7 digits<br>26D |
| <code>mock_{theta_pi_wstp_kernel}.py</code> | Symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF               |

**Core equation:**  $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  
 $W_{26}(n) = \text{Prod}_{\{i=1\}^{26}} [1 + [SSq] \exp(-kappa_i * n/26)]$

#### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`,  
`MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`,  
`uqff_{mock\_theta\_pi\_kernel}.wl`).